



# Introduction to Javascript

1. Client Side Scripting
2. JS Features
3. First Javascript program
4. Language Basics

- Client side scripting basics
  - Client side scripting refers to computer programs which get executed by the browser.
  - It allows to utilize processing power of client machine
  - This avoids unnecessary request-reponse cycles which occupies network bandwidth
- Client side scripting languages
  - Languages like VB Script, JScript, Javascript were used in earlier days
  - Javascript has become the standard client side scripting language now a days
  - Typescript : superset of Javascript with advantage of data types and object oriented features
  - ECMA script: Standard of scripting languages across the web browser

- Basics of javascript
  - It is interpreted language
  - Referred as lightweight programming language.
  - Java and javascript are different languages.
  - Originally developed as as scripting language for Netscape Navigator

- **Lightweight scripting language:** It has a limited set of libraries. Also as it is only meant for client-side execution for web applications
- **Dynamic typing:** In JavaScript we do not have to provide the data type while declaring a variable. Any type of value can be stored in the variable.
- **Object based Lanaguage:** It is referred as object based language because it supprts creating object, maintaining properties and inheriting properties from other objects.
- **Easy syntax:** Similar syntax like C language.

- **Interpreted Language:** JavaScript is an interpreted language which is interpreted by JavaScript interpreter which is a built-in component of the Web browser.
- **Can be easily embedded in web page:** Javascript code can be written in the existing HTML page using script tag which makes HTML page dynamic.
- **Can catch events:** Javascript is designed to react to user actions like button click, checking the checkbox, keeping mouse pointer over certain element, loading of HTML page by calling some user defined function.
- **Case sensitive language:** It follows camel casing notations e.g. getElementById(), setTimeout() etc.

- JS statements should be written in script tag.
- Script tag can be embedded anywhere in the html page and any no of times.
- Note : document is the default object available to javascript code.

```
<html>
<head>
  <title>My First JavaScript
Program</title>
</head>
<body>
  <script>
    // JavaScript code
    document.write("Hello, World!");
    // Print "Hello, World!" on the web
page
  </script>
</body>
</html>
```

- Javascript can be written as external javascript also
  - Create a file with .js extension containing javascript code.
  - <script> tag can be used to link the external JavaScript file.
  - Advantages:
    - It allows for code reuse
    - Clear separation between HTML and javascript code
    - Caching of external javascript file improves speed



- Data types
  - JavaScript does not require you to specify the type of the data contained in a variable
  - It supports following types of values
    - number
    - string
    - null
    - boolean
    - object
    - array
    - Variables

- Variables:
  - Rules for variable naming
  - var, let and const all the keywords are used for declaring variables
  - Comparison between var, let and const
    - Scope
    - Re-declaration allowed?
    - Re-assignment allowed?
    - Initial value
- Operators:
  - All operators are similar like C language
  - Few operators in comparison operators like === and !== are additionally available to compare on type and value both.
  - Special operators like typeof operator is available for knowing the type of existing value

- Conditional statements
  - if..else statement
  - if..else if statement
  - switch statement
- Iterative statements
  - The loops in JavaScript are essentially similar to the ones we are familiar with in C.
  - All three common loop types can be used with JavaScript:
    - while loops
    - for loops
    - do-while loops
  - break and continue can be used in JavaScript loops.

- Defining functions
  - JavaScript functions are used to perform operations.
  - A JavaScript function is defined with the function keyword, followed by a name, followed by parentheses (), followed by pair of curly braces {}.
  - Basic syntax:

```
function functionName([arg1, arg2, ...argN]){  
    //code to be executed  
}
```